

JOHN MARIO MONTTOYA ZAPATA

M.Eng. in Analytics, Postgraduate Specialist in Analytics, Civil Engineer

@ jmmontoyaz13@gmail.com
johnmontoya.vercel.app

+57 310 727 8305
johnma96

john-m-montoya-z-1a375a15b



PROFESSIONAL PROFILE

I am an Artificial Intelligence and Data Science professional with over 6 years of experience, passionate about designing technology solutions and taking them into production environments (MLOps, AIOps). I specialize in developing solutions that improve business processes and decision-making by integrating research and information with technology. I am driven by leading the continuous modernization of organizations, continuous learning, and the democratization of knowledge in multidisciplinary teams.

WORK EXPERIENCE

Data Science Specialist

Protección S.A.

Oct. 2025 – Present Medellín, Colombia

- Designed the RAG layer of a classifier for customer requests and complaints (PQRS) with 150+ categories, using label-agreement routing and LLM tie-breaking: raised automatic coverage from 50% to 75% while maintaining overall system accuracy (~80%).
- Provided technical advisory on the configuration (LLM, rules and harness) of an agentic system on Azure that analyzes the trading desk's chats, emails, calls and meetings to detect fraud and bad-faith negotiations; coverage grew from 50% (chats and emails) to 100% of channels.
- Developed and applied the MLOps framework to deploy models to production on GCP (Vertex AI, Cloud Run, BigQuery, Airflow, GKE) with Docker, Kubernetes, Jenkins and GitLab.
- Took on the technical mentoring of 3 data scientists, supporting them in project scoping, metric definition, tech stack selection and ML model development.

Data Scientist, AI Focus (8 months)

Tuya S.A.

Feb. 2025 – Oct. 2025 Medellín, Colombia

- Designed and implemented generative AI solutions (prompt engineering, agents, vector databases, RAG) for call processing (Speech2Text), information extraction, summarization, chatbots and LLM evaluation agents.
- Developed and optimized ML pipelines that automated model integration into production, and supported the implementation of scalable deployment architectures for ML and generative AI models.

Data Scientist (16 months)

Tuya S.A. (Risk Analytics, Model Risk)

Oct. 2023 – Feb. 2025 Medellín, Colombia

- Led the design and development of credit risk scoring models (probability of default) for revolving products (credit cards), with an AUC-ROC of about 0.80.
- Improved and migrated the credit risk scoring from a no-code workflow tool to Python, along with its on-premise deployment strategy.
- Led the adoption of the MLOps methodology and development best practices with MLflow, DVC, CI/CD, PySpark, Databricks and Scrum.
- Designed, built and maintained the TuyaPy library (OOP), which streamlined code distribution and collaboration across the analytics team.

Data Scientist (18 months)

Comercial Card S.A.S. (PTM)

Mar. 2022 – Sep. 2023 Medellín, Colombia

- Developed predictive models for credit portfolio, fraud and time series (SAGRILAF AML compliance, customer segmentation) and supported their deployment to production with MLOps on Google Cloud.
- Helped build a data warehouse in BigQuery, migrating data with tools such as HEVO.
- Built interactive Power BI reports for financial KPIs and automated calculation processes in Python and KNIME on Google Compute Engine.

Data Analyst (20 months)

Early Warning System of Medellín and the Aburrá Valley (SIATA)

Jul. 2020 – Mar. 2022 Medellín, Colombia

- Analyzed geospatial and temporal environmental data with dimensionality reduction and transformation techniques to support decision-making in early warning and risk management systems.
- Developed distributable Python packages (OOP) for fast reading and analysis of high-frequency time series (20 Hz), supporting risk management.

EDUCATION

Master of Engineering – Analytics

Universidad Nacional de Colombia

Aug. 2023 – Dec. 2025 Medellín

Postgraduate Specialization in Analytics

Universidad Nacional de Colombia

Aug. 2022 – Jul. 2023 Medellín

B.Eng. in Civil Engineering

Universidad Nacional de Colombia

Feb. 2015 – Nov. 2020 Medellín

ACADEMIC EXPERIENCE

Non-invasive diagnosis of the health status of common beans (*Phaseolus vulgaris* L.) in Colombia: an approach based on spectral fingerprinting and artificial intelligence

Master's Thesis

Apr. 2025 – May 2026 Medellín

- Developed a model to detect phosphorus deficiency stress from UAV-captured hyperspectral imagery, selecting 58 spectral bands and vegetation indices.
- Compared ML (logistic regression, LightGBM, XGBoost) and DL models (1D-CNN and 2D-CNN in PyTorch) using a spatial data split to prevent leakage: the 2D-CNN reached a test PR-AUC of 0.96, versus 0.78–0.82 for classical models.
- Ensured experiment reproducibility with MLflow and DVC, and tuned hyperparameters with Optuna.

Spatio-temporal characterization of turbulent flows in the Aburrá Valley

Undergraduate Thesis

Oct. 2019 – Apr. 2020 Medellín

- Performed the statistical analysis of high-temporal-resolution sonic anemometer data (20 Hz).
- Applied frequency analysis techniques (Fourier transform, principal component analysis) to environmental time series.
- Implemented neural networks to model turbulent flow time series.

AWARDS

“Best Undergraduate Theses” Award – Civil Engineering Program (Medellín Campus), 30th edition

Universidad Nacional de Colombia

Apr. 2021 Medellín

“Michel Hermelin Arbaux” Award, 2021 edition

Colombian Academy of Exact, Physical and Natural Sciences

Nov. 2021 Medellín

TECHNICAL SKILLS

Programming: Python and SQL

Machine Learning: scikit-learn, XGBoost, LightGBM, risk scoring, time series

Deep Learning: PyTorch (CNN), Hugging Face

Generative AI: LLMs, RAG, agents, vector databases, LangChain, LangGraph, MCP

Optimization: Optuna, genetic algorithms (hyperparameter search)

Data & MLOps: PySpark, Databricks, BigQuery, Airflow, MLflow, DVC, Docker, Kubernetes, Jenkins, GitLab

Cloud: GCP (Vertex AI, Cloud Run, GKE)

COURSES

Building with the Claude API

Anthropic · Jun. 2026 · Certificate

Introduction to Model Context Protocol

Anthropic · Jun. 2026 · Certificate

Claude Code in Action

Anthropic · Jun. 2026 · Certificate

Generative AI with Google Cloud Specialization

Datapath · Apr. 2026 · Certificate

Databricks and Apache Spark for Big Data: From Zero to Expert

Udemy · Aug. 2025 · Certificate

Advanced MLOps Bootcamp | Machine Learning Operations Hands-on

Udemy · Aug. 2024 · Certificate

LANGUAGES

- Spanish: Native
- English: B1 (Intermediate)

AREAS OF INTEREST

Machine Learning

Deep Learning

Reinforcement Learning

Optimization

Recommender Systems

AI Agents

Credit Risk

Customer Retention

Statistics

Mathematical Modeling

REFERENCES

Jorge Moreno

jhmascus@gmail.com
3014905757

Camilo Ramírez

cramirezg.291@tuya.com.co
3164151048

Mauricio Ramírez

mauricio.ramirez@siata.gov.co
3014678684